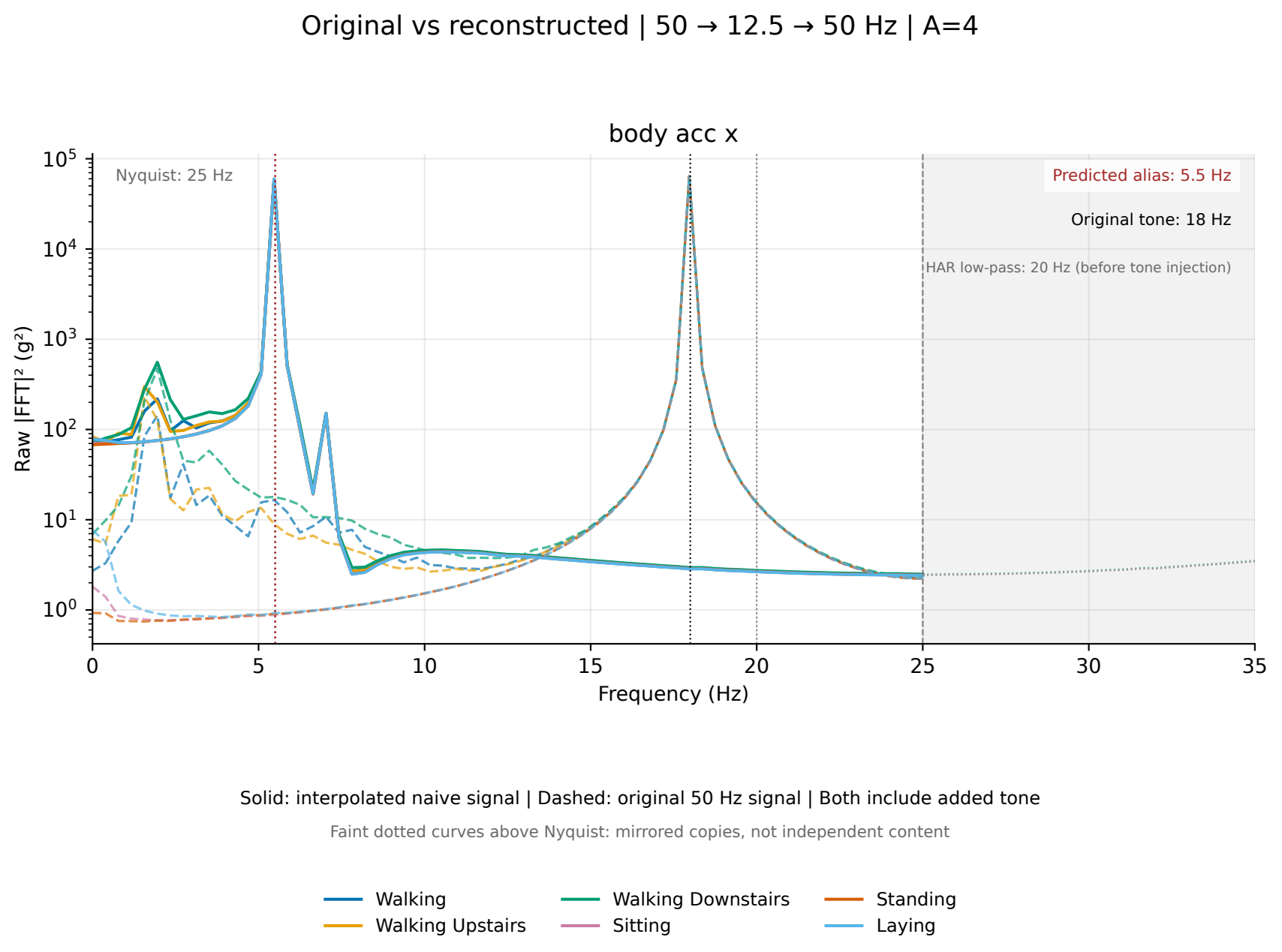
Original vs reconstructed | 50 → 12.5 → 50 Hz | A=4
body acc x
Nyquist: 25 Hz
Predicted alias: 5.5 Hz
Original tone: 18 Hz
HAR low-pass: 20 Hz (before tone injection)
Raw |FFT|² (g²)
Frequency (Hz)
Solid: interpolated naive signal | Dashed: original 50 Hz signal | Both include added tone
Faint dotted curves above Nyquist: mirrored copies, not independent content
Walking
Walking Upstairs
Walking Downstairs
Sitting
Standing
Laying